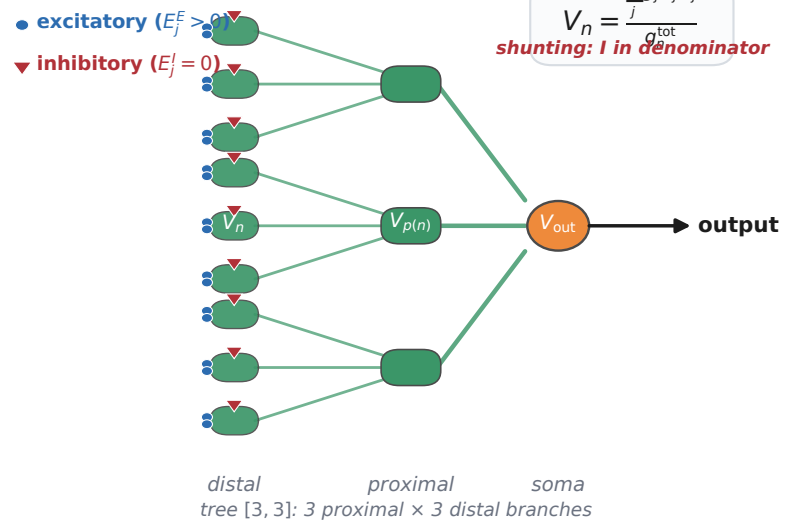


A

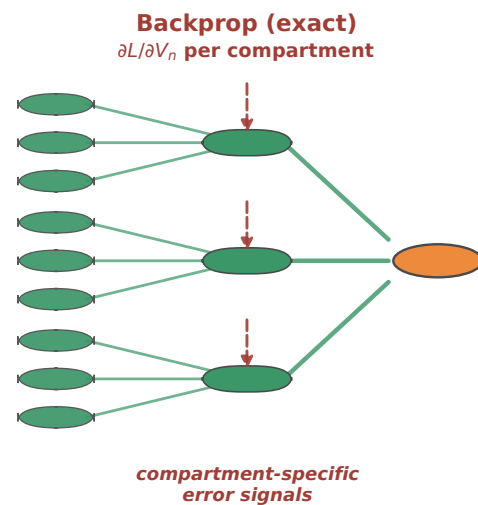
Dendritic neuron with shunting E/I integration

external input

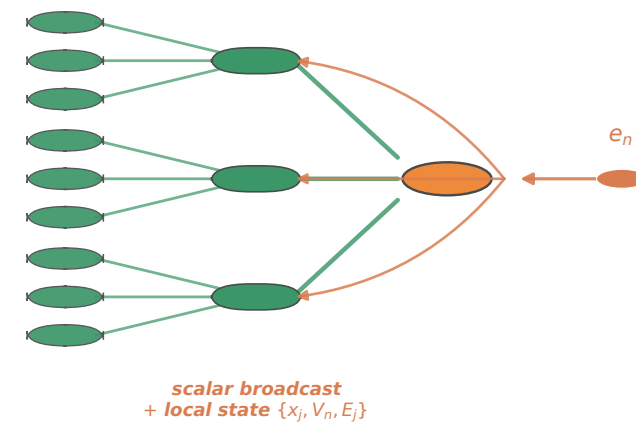


B

Credit assignment: backprop vs. LocalCA



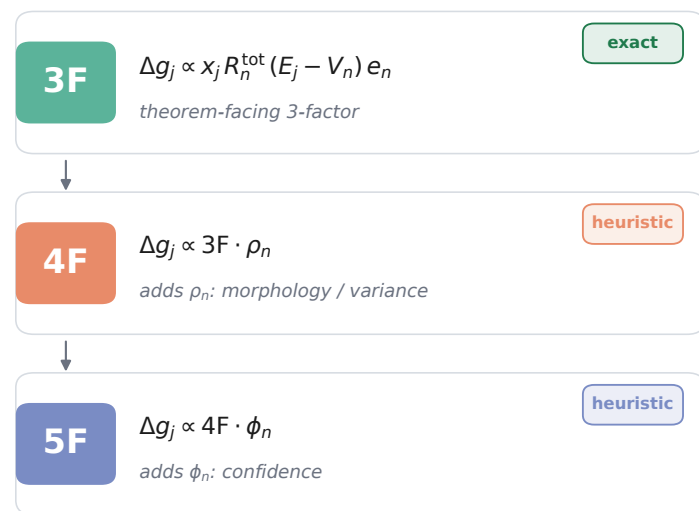
LocalCA (approximate)



Shunting denominator concentrates path gains $\Rightarrow$ scalar e_n is enough

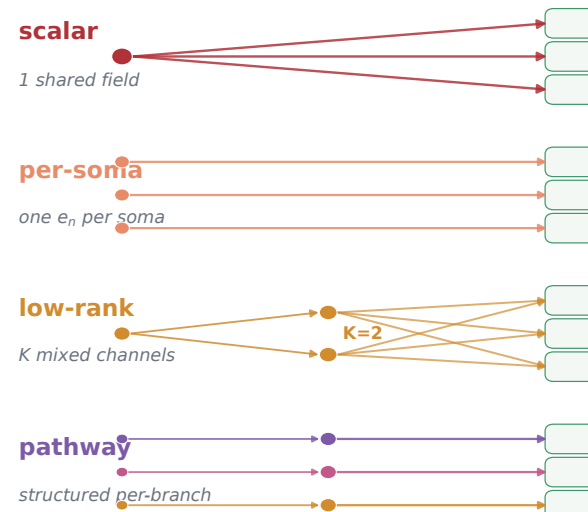
C

Local rule family



D

Broadcast channels for e_n



E

MNIST learning dynamics

